

VERIQO FINAL GAP-CLOSURE AND READINESS REPORT

Assessment date: 25 August 2026

Baseline: Veriqo kernel v7.12.1 R23 / RWC V2 audit baseline

Assessment mode: local engineering qualification and reproducibility review

EXECUTIVE VERDICT

The repository is engineering-qualified for the locally executable scope.

The canonical truth path is wired through source/provenance, rights, evidence, identity, trust, contradiction arbitration, native decision, explanation, durable evidence, certificate, cold replay, and independent verification.

The release is not production-ready. READINESS_MANIFEST.json correctly reports NOT_PRODUCTION_READY: all 58 engineering gates pass, 50 mandatory gates are currently passing in the manifest assessment, and eight mandatory production gates remain externally blocked. No local fixture or synthetic test is treated as equivalent to those external qualifications.

GAP CLOSURE DELIVERED

- Canonical lifecycle.RunUnified path reaches the native execution chain and returns decision, evidence, explanation, certificate, replay, and verification identifiers through the real HTTP gateway.
- Evidence and provenance remain content-addressed and rights-aware; source status is not upgraded to corroborated without independent evidence.
- Trust observations, temporal Bayesian processing, contradiction arbitration, economic consequence, knowledge/fusion components, and native decision are covered by deterministic tests and integration paths.
- Durable WAL/state persistence, crash/restart behavior, ledger anchoring, cold replay, tamper detection, and independent verification are exercised.
- RaftLite, TCP/mTLS transport, flow control, chaos, stress, and soak harness coverage are present and locally executable.
- Gateway integration now reserves an ephemeral loopback port instead of using a fixed port, and captures child-process output on startup failure.
- OS process lifecycle state is synchronized at the process boundary; callers receive snapshots rather than mutable internal process pointers.

VERIFICATION EVIDENCE

Check	Result
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go test ./... -count=1	PASS
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go test -race ./...	PASS
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go vet ./...	PASS
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gofmt -l .	PASS; no output
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./scripts/verify.sh	PASS
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Gateway integration repeated 10x	PASS
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Cold replay HTTP boundary and tamper rejection	PASS
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Eight-blocker fixture qualification	PASS for all eight pipelines
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Readiness manifest generation	Correctly exits 1 with external blockers
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The repository's verify.sh also records the intentionally unavailable checks: network-dependent vulnerability feeds, live SPIRE production attestation, gosec/golangci-lint execution in this sandbox, and OPA/SPIRE external services. Existing evidence files are retained and referenced rather than regenerated into stronger claims.

REMAINING MANDATORY PRODUCTION BLOCKERS

1. HSM/KMS tenancy: production-owned key material, permissions, rotation, revocation, and provider evidence are still required.
2. Authorized live data: contracted source access and live-data rights, freshness, anti-replay, and retention evidence are still required.
3. Multi-region DR: real cross-region/cloud execution with measured RTO/RPO, WAN behavior, traffic-manager cutover, and recovery evidence is required.
4. Independent pentest: an external, scoped penetration test and signed report are required.
5. Scale qualification: the required production-like 100-node execution and measured SLO evidence are not externally qualified.
6. 72-hour soak: a real production-like 72-hour run and signed evidence are still required; the local smoke harness is not a substitute.
7. Production SPIFFE/mTLS: production node attestation (not a demo join-token attestor), SPIRE deployment, and operational revocation evidence are required.
8. Supply-chain vulnerability qualification: a real vulnerability-feed scan is required; local SAST and dependency discovery do not replace it.

RELEASE DISPOSITION

The correct disposition is hold for external qualification, not ?Production Ready?. The source package is suitable for engineering review, controlled qualification, and evidence collection. Promotion requires updating the readiness manifest only after each named external gate has real, content-addressed evidence accepted by the qualification policy.

REPRODUCTION

From the veriqo/ repository root:

```
bash
go test ./... -count=1
go test -race ./...
go vet ./...
test -z "$(gofmt -l .)"
./scripts/verify.sh
go run ./cmd/veriqo-qualification .
go run ./cmd/veriqo-readiness -out READINESS_MANIFEST.json
```

The final command is expected to return exit code 1 until all eight external blockers are qualified. That non-zero result is an intentional safety gate.